CLONE Plugin User Testing Instructions

Thank you for helping us test the CLONE IntelliJ plugin UI. The purpose of this test is to see how understandable, useful, and easy to use the new CLONE UI Panel is when it detects duplicated Java code and suggests a refactoring.

CLONE is an IntelliJ IDEA plugin that detects duplicated code created through copy-and-paste and suggests an Extract Method refactoring. After you test the plugin, please complete the Google Form to help us collect feedback for our research project.

Google Form: https://forms.cloud.microsoft/r/BL5ptNQEjb?origin=lprLink

1. Download the Plugin

Download the CLONE plugin ZIP file from the GitHub repository provided by the research team.

Do **not** unzip the file. IntelliJ will install the plugin directly from the ZIP file.

2. Install the Plugin in IntelliJ IDEA

Open IntelliJ IDEA and open the Java project you will use for testing.

Go to:

Settings → Plugins

In the Plugins panel, click the gear icon and select:

Install Plugin from Disk

Choose the downloaded CLONE ZIP file.

After installing the plugin, restart IntelliJ IDEA if prompted.

3. Configure CLONE

After restarting IntelliJ, go to:

Settings → Tools → CLONE

Under:

Extraction judgements should be performed by

select:

Clone_multiagent

Within “Multi-agent Settings” select Google from the Model Suites dropdown and gemini-3-pro-preview from the Model Selection dropdown. For the Api-Key, enter **AQ.Ab8RN6J1WXunBboxyKEzk5-p8PYqTzUUQDA9y3qeDntLLFSSRQ.**

In order for the tool to work properly, it needs to run on an older version of Java. To do this, navigate to the main menu in IntelliJ and

After saving the settings, restart IntelliJ again if prompted.

4. Run the Test

Once CLONE is installed and configured, open a Java file in IntelliJ IDEA. When ready to begin testing, select the “Build” button from the main menu in IntelliJ (top of the screen) and select “Build Project” from the dropdown.

Copy and paste a block of Java code that creates duplicated logic in the same file or project.

When CLONE detects duplicated code, it should open the UI Panel and guide you through the refactoring review process.

While testing, please pay attention to:

- Whether the clone explanation is easy to understand
- Whether the file locations and line numbers are clear
- Whether the diff preview helps you understand the proposed change
- Whether the AI Trust / validation section helps you decide if the suggestion is reliable
- Whether the buttons and actions are easy to use
- Whether multiple clone cases are shown clearly

5. What to Test in the UI

Please review the main parts of the CLONE UI Panel:

Explanation Section

Check whether the panel clearly explains:

- What type of clone was detected
- Where the pasted code is located
- Where the existing duplicate code is located
- What line numbers are involvedt
- What method or refactoring is being suggested

Diff Preview

Check whether the before-and-after code view makes the proposed refactoring easy to understand.

AI Trust / Validation Section

Check whether the trust score and validation results help you understand whether the refactoring is safe to apply.

This may include:

- Clone Detection
- Refactoring Agent
- Refactoring Usefulness
- Compilation Result
- Test / Behavior Preservation

Multiple Clone Dialog

If more than one clone is detected, check whether each clone is shown separately and whether you can decide what to do with each one.

Buttons and Actions

Check whether the available actions are clear, including:

- Apply
- Edit
- Reject / Cancel
- View Details
- Include / Exclude

Also check whether it is clear why the Apply button is enabled or disabled.

6. Important Notes While Testing

Please test the plugin as naturally as possible. There are no right or wrong answers. We are mainly interested in your experience using the UI.

The plugin may take time to process the pasted code because it uses AI agents and validation steps. Some failures may happen because of API tokens, model limits, or testing environment issues. If an error appears, please mention it in the feedback form.

Add this section under **Important Notes While Testing**:

Token and Code Size Note

Please use a **small and simple duplicated code example** when testing CLONE. The plugin uses AI model calls during clone detection, refactoring, validation, and testing, which can consume API tokens.

A short duplicated code block is better for testing because it is more likely to run successfully and will use fewer tokens.

Example of a good test case:

```
int sum = 0;

for (int i = 0; i < nums.length; i++) {

    sum += nums[i];

}
```

Then paste a similar small block somewhere else in the file.

Please avoid testing with very large methods, long files, or complex code examples at first. If the selected API provider has no available tokens or quota, the plugin may not complete the full workflow. In that case, the UI may show errors, incomplete validation, or failed refactoring attempts. This is expected and should be mentioned in the feedback form.

7. After Testing

After you finish testing the plugin, please complete the Google Form provided by the research team.

Your feedback will help us evaluate:

- How clear the UI is
- Which features are most useful
- Which parts are confusing
- Whether the AI Trust information improves confidence
- Whether users feel comfortable applying the suggested refactoring

Thank you for helping us improve CLONE.